SI: Correcting collection bias in comparative studies of diversity

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1 Rarefaction-based diversity

Rarefaction-based standardization equalizes sampling effort by reducing all collections to the size of the smallest. While widely used, this approach introduces systematic distortions when collection intensity varies across populations. As Figure 1 shows, rarefaction systematically compresses relative richness ratios, particularly among larger populations, especially when sample sizes are small. Crucially, in all conditions rarefaction-based standardization performs worse than taking the uncorrected diversity estimates based on the observed sample sizes.

2 Neutral expectations

To understand basic properties of coverage-based standardization, we use the Wright-Fisher model. In this model, a population of fixed size N evolves in discrete, non-overlapping generations: each new generation is formed by sampling N cultural variants with replacement from the previous generation, so variant counts follow a multinomial distribution. With a small probability μ each sampling event introduces a novel variant not previously seen in the population.

Under the assumption of an infinite allele Wright-Fisher model, the average number of variant types contained in a sample of size n is given by

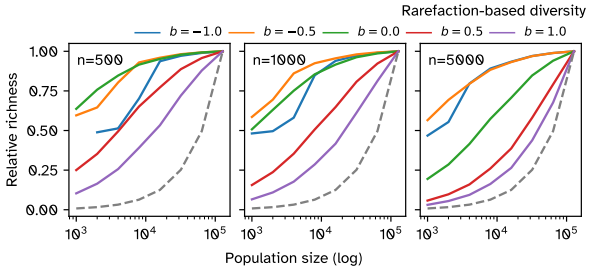
$$\mathbf{E}\{k\} = \sum_{j=0}^{n-1} \frac{\theta}{\theta + j}, \tag{1}$$

and the average coverage of this sample by

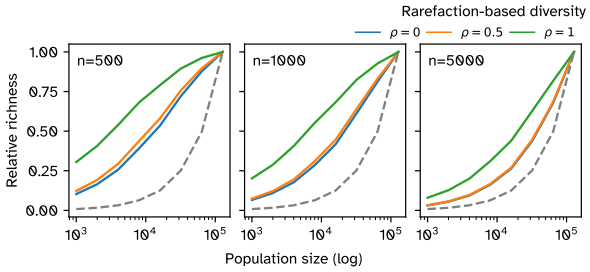
$$\mathbb{E}\{C_n\} = \frac{n}{\theta + n} \tag{2}$$

with $\theta = 2N\mu$.

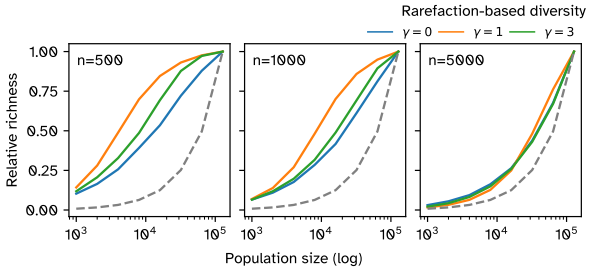
Fig. 2 shows the relationship between sample size n and coverage C_n, as given by (2), for population size $N = 10^4$ and innovation rates $\mu = s{*}10^{-3},\ s = 1, \ldots, 4$. As expected, coverage level is monotonically increasing with sample size: the larger the sample size, the larger the coverage. Naturally, the higher the innovation rate μ_i, i.e., the more variant types are expected in the sample (cf. (1)), the higher the samples size needed to achieve the same coverage. The shaded areas describe the variation of the coverage levels that can be achieved with a given sample size n. For that we simulated 100 Wright-Fisher populations for each considered (N, μ)-combination and calculated their variant-frequency spectrum, i.e., we determined



(a) Population-size bias (b from -1 to 1).



(b) Productivity bias ($\rho \in \{0, 0.5, 1\}$).



(c) Stromer's riddle bias ($\gamma \in \{0, 1, 3\}$).

Figure 1: Rarefaction-based diversity estimates shown as relative richness across population sizes (1,000–128,000 individuals). Each panel applies a different collection bias: (a) population-size bias parameter b; (b) productivity bias with parameter ρ; and (c) Stromer's riddle bias with parameter γ. In all cases, the dashed gray line shows the true underlying richness.

the frequency of every variant type in the population. We then take samples of size n in the range form 10 to 200 and calculate their true sample coverage, C_{true}, by summing the population-level frequencies of all variant types observed in the sample. For low innovation rates, e.g. $\mu = 10^{-3}$, the variation is large for small sample sizes but decreases quickly with increasing n whereas for high innovation rates, e.g. $\mu = 4 * 10^{-3}$, the variation stays similar across the sample sizes considered.

2.1 Accuracy of the Coverage Estimator

In the next step, we analyse the accuracy of the coverage estimator

$$\hat{C}_n = 1 - \frac{f_1}{n}\left[\frac{(n-1)f_1}{(n-1)f_1 + 2f_2}\right] \tag{3}$$

by calculating $\hat{C}_n$ for the set of samples described above and comparing it to the corresponding C_{true}-value. Fig. 3 shows the resulting mean-square error (MSE). Consistent with [1], the MSE is $O(1/n)$. Here $1/n$ is a conservative upper bound and the observed MSE decays faster.

To examine the behavior especially for small sample sizes more closely, Fig. 4 shows the distributions of the $\hat{C}_n$- and C_{true}-values for different innovation rates and sample sizes. For sample size $n = 10$ we observe, especially for high innovation rates, a large proportion of $\hat{C}_n = 0$, i.e., $f_1 = n$. However,

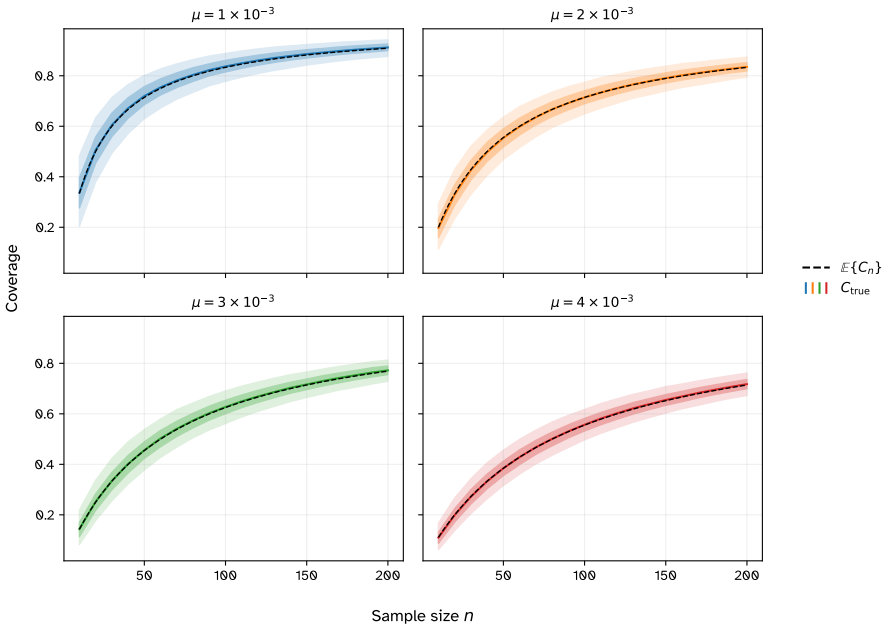


Figure 2: True sample coverage C_{true} as a function of sample size n for Wright–Fisher populations of size $N = 10^4$ under four innovation rates $\mu = s \times 10^{-3}, s = 1, \ldots, 4$. Solid lines show mean coverage across 100 simulated populations; shaded bands denote the 50%, 90%, and full ranges of observed values. Dashed curves indicate the theoretical expectation $\mathbb{E}\{C_n\}$ given by (2). Higher innovation rates lead to lower coverage for a given n and greater variability across replicates.

C_{true} cannot be 0 when $n > 0$. So for a good fraction of the simulations the estimator (3) provides an implausible value. Naturally, the higher the sample size the less pronounced is this effect; more heterogeneous populations require a larger sample size to eliminate the risk of obtaining a $\hat{C}_n = 0$ estimate. As a conservative rule, we recommend that the sample size should be so large that the risk of obtaining an $\hat{C}_n = 0$ estimates is small — for the populations considered here, we would suggest $n \geq 30$. Of course, this risk is very hard to quantify *a priori* in empirical case studies but theoretical studies can provide a rough guideline based on how much heterogeneity researchers expect in their study population. For sample sizes where the fraction of $\hat{C}_n = 0$ estimates is small we see that both C_{true} and $\hat{C}$ distributions become tightly concentrated around $\mathbb{E}[C_n]$ for all innovation rates, indicating that the estimator is stable and unbiased for sample sizes typically encountered in empirical data.

2.2 Accuracy of diversity ratio estimation under coverage-based standardization

The idea of coverage-based standardization is that samples with similar coverage should yield a richness ratio which should approximate the true ratio of the underlying populations. More precisely, when comparing two samples coverage-based standardization proceeds by determining their shared coverage level $\hat{C}_{\max} = \min\{\hat{C}_1, \hat{C}_2\}$, where $\hat{C}_i$ is the coverage estimate (3) for sample i and calculating the ratio of the corresponding richness values $S_1(\hat{C}_{\max})$ and $S_2(\hat{C}_{\max})$.

In the last section we showed that for small sample sizes the realized coverage level can vary substantially around its mean. We now examine how that variability propagates to the shared coverage level $\hat{C}_{\max}$ between two sample and how variability in $\hat{C}_{\max}$ translates into uncertainty in the richness ratio estimate $S_1(\hat{C}_{\max})$ and $S_2(\hat{C}_{\max})$. We aim to provide a rough guideline on sample sizes required for coverage-based standardization to be reliable.

To do so we simulate, as above, Wright-Fisher populations with population size N and innovation rates $\mu = s * 10^{-3}$, $s = 1, \ldots, 4$, determine their population-level richness and draw samples whose *expected* coverage (see (2)) matched a grid of target values $C_{\text{target}} \in [0.1, 0.8]$. On average, both samples should have a very similar coverage level, variations are induced by the sampling procedure. We then

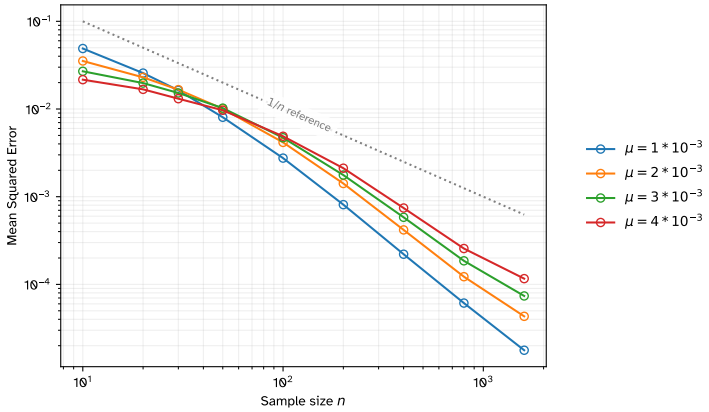


Figure 3: Accuracy of the coverage estimator, $\hat{C}$, across innovation rates and sample sizes. Mean squared error (MSE) of $\hat{C}$ is below and declines approximately in proportion to $1/n$, consistent with theoretical expectations [1]. Each line represents the results for Wright-Fisher models with $N = 10^4$ and different innovation rate μ. Overall, innovation rate μ, and consequently the observed heterogeneity in the population, and sample size n jointly shape the estimator's performance.

quantify the estimation error by comparing $S_1(\hat{C}\text{max})/S_B(\hat{C}_{\max})$ and $S_1^{\text{pop}}/S_2^{\text{pop}}$ where S_i^{pop} defines the richness of the underlying populations i.

Fig. 5 illustrates the accuracy of the diversity estimator $S_1(\hat{C}\text{max})/S_2(\hat{C}_{\max})$. To relate the results to sample size — the only parameter researchers may be able to control — we calculate $n_{\max} = \min(n_1^{\max}, n_2^{\max})$, where $n_i^{\max}$ represents the sample size that results in the shared coverage level $\hat{C}_{\max}$ for sample i.

We binned comparisons by sample size $n_{\max}$. Within each bin we computed the mean absolute relative error. The analysis reveals consistent behavior across innovation rate contrasts. When $n_{\max}$ is small, ($\lesssim 20$ observations), standardized richness ratios are highly unstable and errors are frequently large. Beyond this regime, error declines sharply: once $n_{\max} \gtrsim 30$ errors become much smaller and estimates substantially more stable.

As a rule of thumb, coverage-based standardization may be unreliable when the smallest observed (or adjusted during standardization) sample size is less than 30.

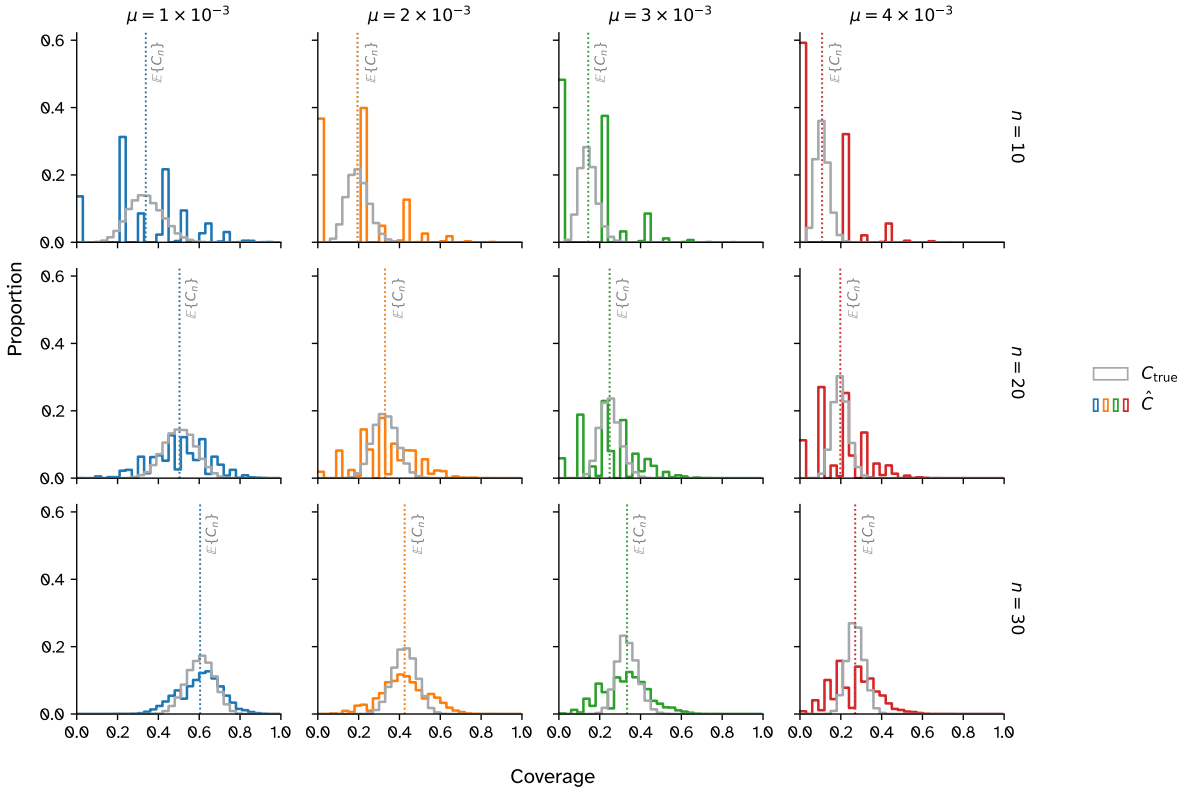


Figure 4: Distribution of true C_{true} and estimated coverages $\hat{C}$ across populations differing in innovation rate μ and sample size n. True coverage C_{true} represents the summed population-level frequencies of all variant types observed in the sample. The theoretical coverage $\mathbb{E}\{C_n\}$ (vertical grey dashed line) gives the expected value according to (2) with $\theta = 2N\mu$.

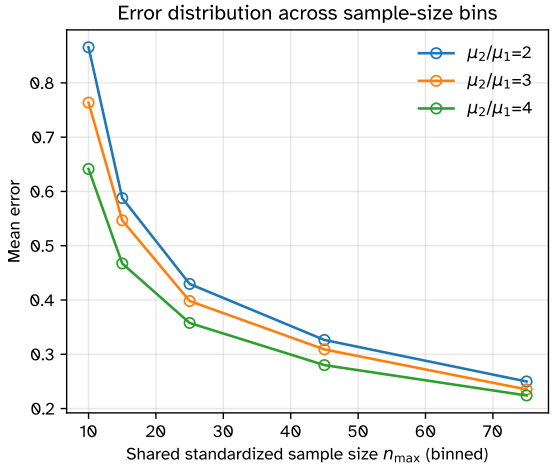


Figure 5: Mean absolute relative error as a function of the achieved standardized sample size $n_{\max}$, grouped into sample-size bins. Despite differences in innovation rate ratios ($\mu_2/\mu_1 \in \{2, 3, 4\}$), all curves show the same pattern: errors fall sharply once ($n_{\max}$) exceeds around 20-30 observations. Relatively modest shared coverage levels ($\hat{C}_{\max} \approx 0.3$) correspond to small standardized sample sizes ($n_{\max} \approx 20 - 30$).

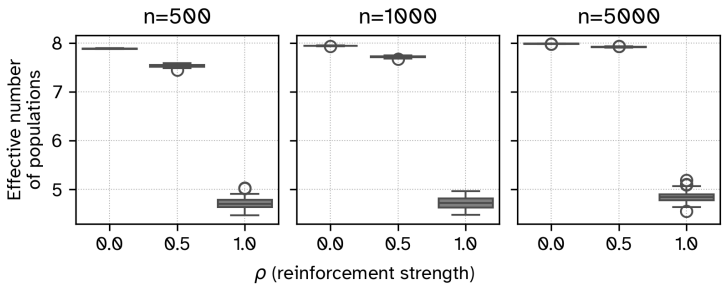


Figure 6: Effect of reinforcement strength (ρ) on the concentration of sampling effort among populations. Each panel shows the *effective number of populations* (Hill number of order 2) after repeated sampling for three total sample sizes ($n = 500, 1000, 5000$). Although mean diversity estimates in the main text appear similar across ρ values, higher reinforcement (larger ρ) leads to a stronger concentration of sampling in a few populations, as reflected by lower effective numbers. Boxes show variation across independent simulation runs.

3 Population-size-bias

For $\beta = 0$, population i is chosen proportional to its size, i.e., with probability $2^{i-1}/\sum_{j=0}^{7}(2^j)$. The average number of times population i is chosen in n draws, i.e. the sample size of population i, is $n_i(n) = n2^{i-1}/\sum_{j=0}^{7}(2^j)$. Consequently, the average number of variants contained in a sample of size n_i is given by (1).

Table 1 shows those expected samples sizes for all eight populations under varying sampling efforts n and the corresponding average number of variants contained in those samples according to (1) and table 2 the average values obtained from the simulation framework.

Table 1: Expected samples sizes n_i for all eight populations of sizes $N_i = 1000 * s^{i-1}$ under varying sampling efforts n and the corresponding average number of variants contained in those samples.

	n_1	n_2	n_3	n_4	n_5	n_6	n_7	n_8
$n = 500$	1.96	3.92	7.84	15.68	31.37	62.74	125.49	250.98
$n = 1000$	3.92	7.84	15.68	31.37	62.74	125.49	250.98	501.96
$n = 5000$	19.60	39.21	78.43	156.86	313.72	627.45	1254.90	2509.80
	$k(n_1)$	$k(n_2)$	$k(n_3)$	$k(n_4)$	$k(n_5)$	$k(n_6)$	$k(n_7)$	$k(n_8)$
$n = 500$	1.5	2.56	4.74	9.13	17.57	35.15	69.63	139.27
$n = 1000$	2.08	3.65	6.85	13.08	25.95	51.29	102.39	204.39
$n = 5000$	3.59	6.55	12.57	24.69	48.90	97.27	194.12	387.77

Table 2: Average samples sizes n_i for all eight populations of sizes $N_i = 1000 * s^{i-1}$ under varying sampling efforts n and the corresponding average number of variants contained in those samples generated by the simulation framework.

	n_1	n_2	n_3	n_4	n_5	n_6	n_7	n_8
$n = 500$	1.95	3.93	7.85	15.62	31.41	62.88	125.47	250.88
$n = 1000$	3.9	7.85	15.63	31.45	62.59	125.4	250.85	502.33
$n = 5000$	19.57	39.09	78.51	156.89	313.86	628.04	1254.22	2509.83
	$k(n_1)$	$k(n_2)$	$k(n_3)$	$k(n_4)$	$k(n_5)$	$k(n_6)$	$k(n_7)$	$k(n_8)$
$n = 500$	1.3	2.39	4.58	9.15	17.46	35.29	69.9	138.97
$n = 1000$	1.94	3.54	6.79	13.45	25.49	51.86	102.66	204.28
$n = 5000$	3.53	6.6	12.7	25.38	48.51	98.67	194.33	388.76

4 Worked example of coverage-based standardization

We illustrate the standardization procedure using two cities from Table 3 below: Leiden and Bruges (Brugge).

Step 1: Compute coverage. For each city, we apply Equation 2 from the main text using the total number of song occurrences n, the number of singletons f_1, and the number of doubletons f_2. For Leiden ($n = 662,\ f_1 = 231,\ f_2 = 69$):

$$\hat{C}_n = 1 - \frac{231}{662}\left[\frac{661 \times 231}{661 \times 231 + 2 \times 69}\right] = 1 - 0.349 \times 0.999 \approx 0.65$$

For Bruges ($n = 201,\ f_1 = 96,\ f_2 = 18$):

$$\hat{C}_n = 1 - \frac{96}{201}\left[\frac{200 \times 96}{200 \times 96 + 2 \times 18}\right] = 1 - 0.478 \times 0.998 \approx 0.52$$

Leiden's sample captures roughly 65% of the underlying melodic repertoire, while Bruges' captures only about 52%.

Step 2: Determine $\hat{C}_{\max}$. To compare the two cities at equal sampling completeness, we set $\hat{C}_{\max}$ to the smaller of the two coverage values: $\hat{C}_{\max} = 0.52$ (Bruges' coverage).

Step 3: Subsample to $\hat{C}_{\max}$. Bruges' sample already has coverage 0.52, so it remains unchanged, with $S_{\text{Bruges}}(\hat{C}_{\max}) = 129$ melodies. Leiden's sample must be rarefied: we draw progressively smaller random subsamples from Leiden's 662 song occurrences, computing $\hat{C}_n$ at each subsample size, until we find the subsample size n^* at which $\hat{C}_{n^*} \approx 0.52$. This yields a coverage-standardized richness of $S_{\text{Leiden}}(\hat{C}_{\max}) = 256$ melodies.

When comparing more than two assemblages (as in Figure 6 of the main text), the same procedure applies. We compute $\hat{C}_n$ for each city and set $\hat{C}_{\max}$ to the minimum coverage across all assemblages. In the cross-sectional analysis, $\hat{C}_{\max} = 0.52$, determined by Bruges (Table 3). Every other city's sample is then rarefied to this common coverage level, and the resulting richness values are those plotted on the y-axis of the right panel of Figure 6.

5 Summary tables

Table 3: Documentation intensity and population metrics across Dutch cities in the 17th century. For each city, we report the number of sources consulted, unique melodies identified, total song occurrences, and estimated coverage of the repertoire. Population size represents historical estimates with associated error margins. Only cities for which 20 or more sources were consulted are included in the dataset. In the cross-sectional analysis, $\hat{C}_{\max} = 0.52$, determined by Bruges (Brugge). σ_N represents a proportional error margin. The absolute uncertainty used in the regression model is $N \times \sigma_N$.

location	# sources	# melodies	# songs	f_1	f_2	C_n	$C_{n\pm}$	N	σ_N
Amsterdam	809	2,003	13,569	733	366	0.95	0.01	149,677	0.05
Antwerpen	180	1,459	7,053	703	222	0.90	0.01	61,982	0.08
Haarlem	92	614	2,795	194	125	0.93	0.01	39,613	0.08
Leiden	65	371	662	231	69	0.65	0.05	45,304	0.06
Hoorn	55	472	1,908	135	123	0.93	0.02	13,906	0.09
Rotterdam	51	188	413	103	38	0.75	0.06	21,906	0.09
Utrecht	42	426	1,163	223	81	0.81	0.03	28,164	0.13
Den Haag	40	222	490	76	94	0.85	0.06	21,064	0.09
Dordrecht	35	129	340	30	66	0.91	0.07	18,023	0.09
Brussel	30	275	877	97	56	0.89	0.03	67,286	0.12
Enkhuizen	30	314	836	158	57	0.81	0.04	17,233	0.09
Alkmaar	28	237	1,056	44	79	0.96	0.03	12,823	0.08
Brugge	28	129	201	96	18	0.52	0.09	30,807	0.07
Delft	26	191	490	76	57	0.85	0.05	20,347	0.09

Table 4: Documentation intensity and population metrics for Amsterdam in the 17th century. For each decade, we report the number of sources consulted, unique melodies identified, total song occurrences, and estimated coverage of the repertoire. In the diachronic analysis, $\hat{C}_{\max} = 0.67$, determined by the 1660 decade. σ_N represents a proportional error margin. The absolute uncertainty used in the regression model is $N \times \sigma_N$.

decade	# sources	# melodies	# songs	f_1	f_2	C_n	$C_{n\pm}$	N	σ_N
1600	38	329	927	163	58	0.82	0.03	56,752	0.05
1610	99	291	1002	122	66	0.88	0.03	78,808	0.04
1620	73	348	1237	120	85	0.90	0.03	92,036	0.04
1630	101	321	779	177	65	0.77	0.04	127,555	0.04
1640	122	572	2597	210	124	0.92	0.02	142,153	0.05
1650	85	581	2229	218	154	0.90	0.02	172,644	0.05
1660	80	378	725	236	79	0.67	0.05	187,175	0.05
1670	69	420	1191	140	136	0.88	0.03	190,929	0.05
1680	92	692	2066	289	168	0.86	0.02	198,540	0.06
1690	50	409	816	208	100	0.75	0.05	199,624	0.06

References

[1] Herbert E. Robbins. “Estimating the total probability of the unobserved outcomes of an experiment”. In: *The annals of Mathematical Statistics* 39 (1 1968), pp. 256–257.